

ECON 21030: Honors Econometrics

Regression Writ Large

Spring Quarter 2024

Linear models and regression

We have now covered how to model the distribution of random variables and learn about those distributions from random samples. Sometimes in econometrics we are interested in learning about the distribution of a single random variable, but more frequently we want to know how two or more random variables are related. We might ask, for example, whether individuals who have more schooling tend to earn more money on average, whether police are less likely to stop a car when the driver is white, whether prices are higher for goods with import restrictions, and so on.

The workhorse tool for modeling the relationship between two random variables is the simple [linear model](#). For random variables Y and X , we write the linear model as:

$$Y = \alpha + \beta X + U$$

where U is a new random variable defined as $Y - \alpha - \beta X$. It is often called the [residual](#). Y is typically referred to as the [outcome](#) variable, while X is called a [covariate](#) (for reasons that will become clear shortly). I want you to be careful to note that this is a *statistical* model—all we have done is write down an accounting identity connecting Y , X and U that would hold for any α and β . Though Y and X may exist in the real world and measure real things, U is just defined to satisfy the linear restriction. When and how we can interpret β as telling us something about the *causal* effects of X on Y will come later.

We can write down a linear model for any α, β . Some α and β are special, and among the most special are the α and β implied by [linear regression](#). In linear regression, these parameters are pinned down by two extra restrictions: $E[U] = 0$ and $E[UX] = 0$. In fact, these restrictions imply:

$$\begin{aligned} E[U] &= E[Y - \alpha - \beta X] = 0 \\ &\rightarrow \alpha = E[Y] - \beta E[X] \\ E[UX] &= E[(Y - \alpha - \beta X)X] = 0 \\ &\rightarrow \beta E[X^2] = E[YX] - \alpha E[X] \\ \beta(E[X^2] - E[X]^2) &= E[YX] - E[Y]E[X] \end{aligned}$$

Whenever $E[X^2] - E[X]^2 > 0$, this system implies a unique value for β given by:

$$\beta = \frac{E[YX] - E[Y]E[X]}{E[X^2] - E[X]^2} = \frac{Cov(Y, X)}{Var(X)}$$

while α will be given by $E[Y] - \beta E[X]$. The parameter α is called the **intercept** or **constant**, while β is called the **coefficient** on X .

Exercise: What happens when $Var(X) = 0$. Is the linear regression model still defined?

Linear regression turns out to solve a very particular problem. It seeks to find the (α, β) that minimize the expected squared residual, i.e. $\min_{\alpha, \beta} E[U^2]$. To see why, note that the first order conditions for minimizing $\min_{\alpha, \beta} E[(Y - \alpha - \beta X)^2]$ imply:

$$\begin{aligned} 2E[Y - \alpha - \beta X] &= 0 \rightarrow E[U] = 0 \\ 2E[X(Y - \alpha - \beta X)] &= 0 \rightarrow E[XU] = 0 \end{aligned}$$

This is why linear regression is said to solve a **least squares problem**. The parameters defined by these moment conditions make the expected square of U as small as possible. Since $E[U] = 0$, this implies that linear regression model also minimizes **residual variance**.

Linear regression is also called the **population projection** of Y onto X . The projection part of this name comes from the geometric interpretation of least squares, which views the model as projecting Y on to the space spanned by X .

The population part of this name comes from the fact that we are dealing here with random variables Y and X . In the next lecture we will discuss regression in samples, where we only observe a finite number of observations and where we will define sample projections as well. The word population here thus helps to distinguish what will ultimately be an estimand (i.e. (α, β)) from an estimate obtained in a sample (i.e. $(\hat{\alpha}, \hat{\beta})$).

When X and Y are both scalar random variables, we call this problem **bivariate regression**. Of course, we can also consider **multivariate regression**, which is the model defined by:

$$\begin{aligned} Y &= \alpha + \mathbf{X}'\beta + U \\ E[U] &= 0, \quad E[\mathbf{X}U] = 0 \end{aligned}$$

where $\mathbf{X}$ is a k -dimensional random variable and β is a k -dimensional parameter. The restriction that $E[\mathbf{X}U] = 0$ means that $E[X_k U] = 0$ for each component X_k of X . That is, it is a system of restrictions. I'm going to drop the bold notation going forward, but keep in mind that X is still a random vector.

You will often see X and β broken out into each component, e.g., $Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + U$, when it is helpful to distinguish between the coefficients on different components of X . It is also common to write multivariate regression partitioning X into two sets of random variables, e.g., $Y = \alpha + X'\beta_1 + W'\beta_2 + U$, where $\dim(X) + \dim(W) = k$. All of these are versions of the same thing.

Since $\alpha = \alpha X_0$, where X_0 is random variable that always takes on the value of 1, it is also common to write the multivariate regression model even more compactly as $Y = X'\beta + U$, where the first component of X is just a one (i.e., the degenerate random variable X_0) and the first component of β is α . Written this way, the restriction that $E[XU] = 0$ embeds the restriction that $E[U] = 0$ in the first component.

Just as in the bivariate case, the restrictions on U can bin down α and β in this case.

$$\begin{aligned} E[XU] &= E[X(Y - X'\beta)] = 0 \\ \rightarrow E[XX']\beta &= E[XY] \end{aligned}$$

Notice that $E[XY]$ is a k -dimensional random vector and XX' is a k by k random matrix. Thus whenever $E[XX']^{-1}$ exists, we can solve for a unique β that satisfies our moment restrictions. In other words, β in the multivariate regression problem is *defined* as $E[XX']^{-1}E[XY]$. The existence of this inverse will happen whenever $E[XX']$ is full rank, which requires that the components of X be linearly independent. A sufficient condition is that there is no vector c such that $Pr(X'c = 0) = 1$, ruling out settings where some components of X and linear functions of other parts. We cannot, for example, find a unique β when $X = (X_1, 2 \cdot X_1)'$, for example. The rank condition is why [colinearity](#) among covariates is problematic, an issue we will return to later.

Exercise: What happens when $E[X'X]$ is *not* full rank. Is the multivariate regression model still defined?

The random variable defined by $\tilde{Y} = X'\beta$ is called the [fit](#) or [predicted values](#) from this regression. It is sometimes useful to ask how good the “fit” here is, meaning how correlated are Y and $\tilde{Y}$. The [population \$R^2\$](#) (pronounced “r-squared”) is defined as:

$$R^2(\beta) = 1 - \frac{Var(Y - X'\beta)}{Var(Y)} = 1 - \frac{Var(U)}{Var(Y)}$$

I’m using “population” here again to remind you hear that we are talking about random variables and the coefficient β from a population projection, not something that happens in sample. We will encounter sample R^2 later.

Clearly when the fit is very good, in the sense that U is usually very small, R^2 will be high. In fact, you can show that R^2 is the squared correlation between $\tilde{Y}$ and Y . Proving this fact serves as a useful exercise to reinforce some of the properties of liner regression we have introduced above:

$$\begin{aligned} R^2(\beta) &= 1 - \frac{Var(Y - X'\beta)}{Var(Y)} = \frac{Var(Y) - Var(Y - X'\beta)}{Var(Y)} \\ &= \frac{2Cov(X'\beta + U, X'\beta) - Var(X'\beta)}{Var(Y)} \\ &= \frac{Var(X'\beta)Var(X'\beta)}{Var(Y)Var(X'\beta)} = \frac{Cov(Y, X'\beta)^2}{Var(Y)Var(X'\beta)} \end{aligned}$$

We used the fact that $Cov(X'\beta, U) = 0$ repeatedly here. This of course has to be true because $Cov(X'\beta, U) = Cov(\beta'X, U) = \beta'E[XU] = 0$ by construction. This is why, for example, $Var(X'\beta) = Cov(X'\beta, X'\beta) = Cov(X'\beta + U, X'\beta) = Cov(Y, X'\beta)$.

Remember, we are not making any claims about the casual effects of X on Y . We have simply written down a linear model and two restrictions that imply (under some conditions) a unique set of parameters. Under these conditions, these parameters *always* exist for any two random variables / vectors Y and X , whatever they may be. We are free to regress total incidents of piracy on global average temperatures, IQ scores on hours spent watching star wars Christmas specials, income on reported fondness for dogs, whatever.

In other words, this is a purely *statistical* model. A positive β , or a negative one, or a large R^2 does not have any deep meaning. These are simply properties of the relevant random variables and their covariances. In particular, you should not assume that a large R^2 makes a regression “good.” Whether and when a regression is good will depend on the question, the data, and the

assumptions you make, and very often important causal questions have nothing to do with the quality of model fit.

So why is regression so useful? And why would we ever want to estimate β ? It turns out that regression has a bunch of useful properties that imbue β with some special statistical interpretations, making it a particularly convenient way to summarize relationships between X and Y . These properties also make regression the right tool when we're equipped with sufficient extra information to causal claims. We'll cover these special interpretations next.

Conditional expectation functions

Recall from our first lecture the definition of the conditional expectation function (CEF), or $E[Y|X]$. The CEF is the random variable defined by a transformation of X using the function $\mu(x) = E[Y|X = x]$. Now suppose that this conditional expectation function is linear, meaning that $\mu(x) = x'\beta$. Then it turns out that linear regression delivers the CEF.

To see why, notice that for any conditional expectation function, we can always write $Y = E[Y|X] + U$, where U has the following properties:

$$\begin{aligned} E[U] &= E[Y - E[Y|X]] = E[Y] - E[E[Y|X]] = 0 \\ E[U|X] &= E[Y - E[Y|X]|X] = E[Y|X] - E[Y|X] = 0 \end{aligned}$$

Both lines follow from the law of iterated expectations. The second line implies that $E[XU] = 0$ because $E[XU] = E[E[XU|X]] = E[XE[U|X]] = 0$. Thus when $E[Y|X] = X'\beta$, we can write a model such that $Y = X'\beta + U$ with $E[U] = E[XU] = 0$. In other words, claiming that the conditional expectation function is linear in β *implies* β is the population projection coefficient on X .

Exercise: How are the assumptions that $E[U|X] = 0$ and $E[XU] = 0$ related? Which is stronger?

Another way to see the relationship between linear CEFs and regression is use the definition of the population projection coefficient, which I'll call β_{reg} to distinguish it from the β in the CEF, when $E[XX']$ is invertible. In this case, we have:

$$\begin{aligned} \beta_{reg} &= E[XX']^{-1}E[XY] = E[XX']^{-1}E[E[XY|X]] \\ &= E[XX']^{-1}E[XE[Y|X]] = E[XX']^{-1}E[XX'\beta] = \beta \end{aligned}$$

Thus, again, regression and the CEF coincide. So if we are interested in the CEF, which is a very useful way to describe how Y varies with X , and have reason to believe the CEF is linear, then regression puts us in business.

It turns out that CEFs are also special in the sense that they are the best predictors of Y given X in a mean squared error sense. That is:

$$\mu(x) = \operatorname{argmin}_g E[(Y - g(X))^2]$$

The proof is straightforward and another good illustration of some of the ideas we've covered already. Suppose that some g minimizes $E[(Y - g(X))^2]$. Then:

$$\begin{aligned} E[(Y - g(X))^2] &= E[((Y - \mu(X)) - (g(X) - \mu(X)))^2] \\ &= E[(Y - \mu(X))^2] + E[(g(X) - \mu(X))^2] - 2E[(Y - \mu(X))(g(X) - \mu(X))] \\ E[(Y - \mu(X))(g(X) - \mu(X))] &= E[E[(Y - \mu(X))(g(X) - \mu(X))|X]] \\ &= E[(g(X) - \mu(X))E[(Y - \mu(X))|X]] = 0 \end{aligned}$$

It follows that $E[(Y - g(X))^2] > E[(Y - \mu(X))^2]$ unless $g(X) = \mu(X)$.

This tells you that when the CEF is linear and therefore coincides with the population projection, regression is also giving us the best predictor of Y given X in mean squared error. There is *no* better predictor of Y given X under squared loss.

There are some special cases when the CEF is linear (e.g., jointly normal random variables), but this need not generally be the case. So what if the CEF is *not* linear? Then we know that regression will not give us the CEF, and that another function (specifically, the CEF itself) is a better predictor of Y given X than the $X'\beta$ under squared loss. Can we say anything about the value of regression in this case?

The answer is yes. First, we have already shown that among *linear* predictors of Y , regression is the best in terms of mean squared error. For this reason regression is often referred to as the **best linear predictor** (BLP). So if we want to restrict the class of g to be linear, regression will be the winner.

Second, the proof of why the CEF minimizes mean squared error shows that:

$$E[(Y - g(X))^2] = E[(Y - \mu(X))^2] + E[(\mu(X) - g(x))^2]$$

Since linear regression minimizes $E[(Y - g(X))^2]$ among the class of linear g , it must also minimize $E[(\mu(X) - g(x))^2]$. Thus linear regression is also the MSE-minimizing linear predictor of the CEF. That is, regression is the **best linear approximation** to the CEF. Even when the CEF is potentially highly non-linear, we may therefore be interested in linear regression as useful approximation either because the non-linearity is too complex or irrelevant to our interests, or because don't know what form the non-linearity might take.

Since the CEF is best we can do in terms of mean squared error for predicting Y given X , it turns out we don't even need Y to define regression. In fact, β is also the solution to a regression of $\bar{Y} = E[Y|X]$ on X . We already showed a version of this result above, where by iterated expectations we found that $\beta = E[XX']^{-1}E[XY] = E[XX']^{-1}E[XE[Y|X]]$. This should make more sense now that you know that regression is just the best linear fit to the CEF of Y given X itself—that is, regression “fits the means.” This also illustrates that knowing the conditional means of Y given X is sufficient to compute β —other moments (e.g., conditional variances) are irrelevant. Sometimes in practice you will encounter settings where you are missing the “micro-data” and only have access to group means and will need to make use of this result.

Linearity may seem somewhat restrictive. Keep in mind, however, that regression can accommodate models that are linear in *functions* of X . If we think the correct model for $E[Y|X]$ is $\mu(X) = \alpha + \beta_1 X + \beta_2 X^2$, for example, we can consider multivariate regression with regressors $(1, X, X^2)'$.

Likewise, if we think the CEF involves [interactions](#), such as $E[Y|X] = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$, we can consider linear regression of Y onto $(1, X_1, X_2, X_1 X_2)'$. In each case β will be uniquely defined so long as our rank condition holds. And in each case regression will deliver the (in this case non-linear) CEF.

Of course, in order to construct the appropriate regressors in this case we had to know the [functional form](#) of the CEF—i.e., that it is linear in X and X^2 , for example, but does not depend on X^3 .¹ There are some special cases, however, when we *know* that regression recovers the CEF. A simple example is when X is a binary random variable. In this case, the CEF has two points of support: $E[Y|X = 0]$ if $X = 0$ and $E[Y|X = 1]$ if $X = 1$, i.e., two simple conditional means. In this case, one can show (as you will on your first homework) that a linear regression of Y on X will yield:

$$\begin{aligned}\alpha &= E[Y|X = 0] \\ \beta &= E[Y|X = 1] - E[Y|X = 0]\end{aligned}$$

β is a simple difference in means. If Y were income for example and X were an indicator for being male, β would just tell us the male-female gap in income. Although we could have also computed this difference in means directly from the data, regression is often a helpful way to measure these means when you want to add some bells and whistles, such as non-standard inference or adjustments for other covariates (more on this later).

The binary case is straightforward, but the same ideas generalize to higher dimensions. If X were discrete with k points of support, we could also recover the CEF using regression by using as regressors k [indicator functions](#) or [dummy variables](#), i.e., $(1\{X = 1\}, 1\{X = 2\}, \dots, 1\{X = k\})$. Regression on this multivariate set of binary random variables would also recover the CEF.

If X were a set of several discrete random variables, we can still recover the CEF so long as the regressors are [saturated](#). Saturation means that there is unique element of β for all possible values X can take on. Saturated regressions always recover the CEF because in these settings the CEF consists of a discrete random variable with finite support. If you can dummy-out the points of support, you will get back the CEF.

As a simple example, suppose a Y is income and X consists of two random variables, a binary indicator for being male and a binary indicator for working in finance. A regression of Y onto the these two random variables and a constant alone is *not* saturated. There are only three parameters, but X can take on four unique values (male in finance, male not in finance, female in finance, female not in finance). Including an additional regressor that is the interaction of the these two binary random variables (i.e., an indicator for being male and working finance) would make the regression saturated.

As in the binary case, these regression coefficients have interpretations as differences in means. Your second homework will walk you through some simple examples to help you get practice constructing and interpreting saturated sets of regressors.

¹We will cover non-parametric estimators of the CEF later in the course.

Regression algebra

Let's go back to a world with outcome Y and two sets of regressors X_1 and X_2 . Suppose I am interested in the projection of Y onto both regressors:

$$Y = X_1'\beta_1 + X_2'\beta_2 + U$$

Earlier we showed that in the simple bivariate case β has simple interpretation related to the scaled covariance between Y and X . What about here? The [Frisch-Waugh-Lovell](#) (FWL) theorem provides a insightful way to make progress on this question. FWL says that:

$$\begin{aligned}\beta_1 &= E[\tilde{X}_1\tilde{X}_1']^{-1}E[\tilde{X}_1Y] \\ \beta_2 &= E[\tilde{X}_2\tilde{X}_2']^{-1}E[\tilde{X}_2Y]\end{aligned}$$

where $\tilde{X}_1 = X_1 - (X_2'E[X_2X_2']^{-1}E[X_2X_1'])'$ and $\tilde{X}_2$ has been defined analogously. You will prove a version of this on your homework.

In words, FWL says that we can recover β_1 by first [residualizing](#) X_1 on X_2 , then regressing Y on these residuals ([exercise: does it matter whether we residualize \$Y\$ in the same way? Why or why not?](#)). Residualizing means to project each element of X_1 onto X_2 and compute the residuals—i.e., $X_1 = X_2'\gamma + \tilde{X}_1$ (note: what is the dimension of γ and $\tilde{X}_1$ here?). The process of residualizing X_1 on X_2 is also known as [partialling out](#) X_2 .²

In particular, when X_1 is a scalar random variable, we can represent β_1 as $Cov(Y, \tilde{X}_1)/Var(\tilde{X}_1)$. This is a very useful way to go from a big, multivariate linear model into a simple bivariate one. We just have to first residualize the regressor of interest on the other covariates, then work with these residualized variables. In particular, for any k -dimensional vector of regressors X , we can represent the coefficient on X_k as $Cov(Y, \tilde{X}_k)/Var(\tilde{X}_k)$, where $\tilde{X}_k$ is X_k after residualization on the other $k - 1$ covariates. Of course, as in my leading example from above FWL also works when we only want to residualize *blocks* of covariates.

As an aside, FWL also gives some simple intuition for why perfect collinearity is a problem in regression. If component k is a linear combination of the remaining $k - 1$ components of X , then there is some vector c such that $X_k = X_{-k}'c$, where X_{-k} is the vector of random variables excluding X_k . But then $\tilde{X}_k = 0$ everywhere, since c must be the coefficient on X_{-k} in the regression of X_k on X_{-k} because it minimizes the residual sum of squares. $Var(\tilde{X}_k) = 0$ therefore, which makes the coefficient on $\tilde{X}_k$ not unique defined.

You will find FWL particularly useful when you want to make a graph of the relationship underlying β_1 , which is a lot easier to do in two dimensions than $k + 1$, or when you are only interested in one X_k (i.e., some kind of “treatment”) but want to adjust / control for a variety of other factors. FWL allows you to account for other regressors and then effectively ignore them. Sometimes FWL tricks provide computational speeds ups that make life easier when working with actual data as well.

The FWL logic also underlies the famous [omitted variables formula](#), which is a neat way to relate coefficients from projections with nested subsets of regressors. Suppose I am interested the

²Note that one of X_1 or X_2 could include a constant here, but that is not necessary for FWL to hold. We could have also written in a constant term explicitly as $Y = \alpha + X_1'\beta_1 + X_2'\beta_2 + U$, which would have changed the expressions for β_1 and β_2 slightly ([exercise: how?](#)).

regression of Y onto two scalar random variables:

$$Y = \alpha_0 + \beta_1 X_1 + \beta_2 X_2 + U$$

But I only have access to the coefficients from the bivariate regression of Y on X_1 :

$$Y = \alpha_1 + \beta_s X_1 + V$$

How is β_s related to β_1 ? In this simple bivariate case, we know that:

$$\begin{aligned}\beta_s &= \frac{Cov(Y, X_1)}{Var(X_1)} \\ &= \frac{Cov(\beta_1 X_1 + \beta_2 X_2 + U, X_1)}{Var(X_1)} \\ &= \beta_1 + \beta_2 \frac{Cov(X_1, X_2)}{Var(X_1)}\end{aligned}$$

Notice that $Cov(X_1, X_2)/Var(X_1)$ is the coefficient on X_1 in a regression of X_2 on X_1 . Thus β_s , which I have named because the regression without X_2 is often called the [short](#) regression, is equal to a β_1 , the coefficient on X_1 in the [long](#) regression, plus β_2 times the coefficient on X_1 in a regression of X_2 on X_1 .

In general, β_1 and β_s thus will only coincide when X_1 and X_2 are uncorrelated. In that (special) case, adding X_2 to the regression does not affect the coefficient on X_1 . Of course, we already knew this to be true from FWL logic, since if $Cov(X_1, X_2) = 0$ then $\tilde{X}_1 = X_1$, so residualizing has no effect.